

MENGYU HAN

Software Engineer

Beijing, China | +86 157 9070 9643 | seaseahan79@gmail.com

[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Software engineer with **5+ years at SLB**, delivering cloud applications and scientific visualization with **Angular, C#/.NET, Three.js and GLSL**. Rapidly learns and applies new technologies and domain knowledge to deliver large-scale reservoir rendering and a patented site-selection algorithm. Communicates complex requirements across domain experts and global engineering teams; led a **3-person team** and trained **20 engineers**.

PROFESSIONAL EXPERIENCE

SLB (Schlumberger) | Beijing, China

2021 - Present

Prism | Geological Interpretation and Reservoir Visualization

Cloud platform for geoscientists to interpret well data and explore 3D reservoir models.

Tech stack: Angular, TypeScript, RxJS, Three.js, WebGL/GLSL, C#/.NET, Module Federation, AKS, OpenTelemetry, Prometheus, Grafana

- Made multi-million-cell reservoir models responsive in the browser by replacing geometry-heavy rendering with **Three.js/GLSL GPU volume ray casting** and integrating chunked streaming with encrypted browser caching. Improved rendering from **0-15 to sustained 60+ FPS** on the same data and hardware; reduced loading for **~9M elements from ~3-5 min to ~30-40 s initially / ~5 s cached**, depending on network conditions. A related **patent application is in progress**.
- Led a **3-person team** to deliver **Dip Picking**, a key capability for winning customers. Enabled geologists to measure the tilt and direction of underground rock layers from well data through interactive editing and automated geological calculations, closing a longstanding product gap. The feature **passed 30+ real-time drilling tests worldwide**, earning customer trust and helping **SLB Digital gain a competitive advantage in shale oil**.
- Built reusable **Module Federation microfrontend plugins**, enabling product teams to integrate features and iterate faster.
- Developed **C# REST APIs** and contributed to frontend and backend observability on AKS with OpenTelemetry, Prometheus and Grafana for runtime monitoring.
- Coordinated with external vendors and internal engineering teams to resolve **13+ integration issues**, guided contractor implementation, and delivered two technical training sessions for **20 engineers across Beijing and Houston**.

CarbonSCORE 2.0 | Carbon Storage Site Selection

Decision-support application for comparing and ranking potential carbon-storage sites.

Tech stack: Angular, RxJS, AG Grid, C#/.NET, AHP, TOPSIS

- Researched **10 decision-making methods** and co-developed an **AHP + TOPSIS** algorithm with domain experts, resolving a product-delivery blocker. Reduced the growth in required user comparisons from quadratic to linear ($O(n^2) \rightarrow O(n)$) as site count increases. The algorithm was adopted in CarbonSCORE 2.0; named co-inventor on [U.S. Patent 12,404,748](#).
- Built the **full-stack prototype** that guided product development: C# REST APIs for site scoring, criteria weights and consistency checks, plus an Angular/RxJS interface with customized AG Grid components.

Cybertron | Cloud Application Development

Cloud applications for subsurface data management and visualization.

Tech stack: Angular, NgRx, RxJS, C#/.NET, Go (Golang), gRPC, OpenAPI, Docker, Kubernetes, GCP

- Delivered Angular data-management and visualization interfaces with reusable components, NgRx state management and RxJS asynchronous data flows.
- Built **C# APIs** to import measurements of underground rock layers into a cloud platform and track processing status; developed **Go services** to handle multiple tasks at once, used **gRPC** for service-to-service communication, and documented the APIs.
- Supported C#/Go service deployment and troubleshooting on **GCP Kubernetes**; developed Taiko + Gauge UI automation and API/service tests to validate application workflows.

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ADDITIONAL EXPERIENCE

Enterprise Application Modernization | SLB

Modernization and operation of internal business applications.

Tech stack: Angular, .NET Framework, .NET Core, SQL, SQL Server, Oracle, IIS, Windows Server, HTTPS/TLS

- Coordinated a **3-person team** to upgrade **Angular 7 to 16**, migrate **.NET Framework to .NET Core**, upgrade Windows Server and renew HTTPS certificates; resolved compatibility and deployment issues as part of ongoing application maintenance.
- Wrote **SQL queries** to extract business data from SQL Server and Oracle databases and support management decision-making.

AWARDS AND RECOGNITION

- **Out of Ordinary (O2) Award** | Q2 2026.
- **Out of Ordinary (O2) Award** | Q4 2024.
- **Collaboration in Innovation, Q1 2026** | Cross-center 3D Gaussian Splatting integration with Spark.js and Three.js.
- **Asian Hackathon - Best Collaboration Award, 2025 (team award)** | Co-developed an **AI document-splitting and search system with RAG (retrieval-augmented generation)** for SLB's quality inspection center, reducing completion time for comparable tasks by **70%**.
- **Gesture-interaction projects, 2024:** Euro Hackathon **1st Place** (54 submissions); BGC Innovation Carnival **Best Presentation** (102 submissions).
- **BGC Coding League - 3rd Place, 2025** | Frontend development competition; 24 teams.
- **BGC Bootcamp - Best Classroom Performance Award, 2022.**

VOLUNTEER AND COMPANY EVENT PROJECTS

SLB100 Family Day | Gesture-Controlled 3D Game | [Demo](#) | [GitHub](#)

React, Three.js, MediaPipe. Built a camera-controlled 3D game for SLB's centennial event, combining gesture-sequence recognition, timed holds and animated challenges.

2024 IDPwD | Sign Language Game | [Demo](#) | [GitHub](#)

Angular, MediaPipe. Designed and built a webcam-based sign-language game for SLB's International Day of Persons with Disabilities, helping participants explore accessible communication.

WhatIf Studio | SLB100 Family Day | [UI preview](#) | [GitHub](#)

Next.js, Python, InsightFace/inswapper. Built an AI portrait experience for SLB's centennial event, integrating face uploads, generation and a portrait gallery. Public preview shows the interface.

PATENTS AND PUBLICATIONS

U.S. Patent 12,404,748 | Co-inventor | Granted Sep 2025

[Carbon Sequestration and Storage Site Selection and Usage](#)

IEEE ICSSSM 2019 | Graduation thesis | Xiukun Zhao, **Mengyu Han**, Qianzhou Deng, Kelei Xue
[Decisions on supply and demand of bicycle-sharing service based on cobweb model](#)

EDUCATION

Washington University in St. Louis | Master of Information Systems Management, 2019

Tianjin University of Finance and Economics | B.S. Management Science, 2018